

HINTS AND SOLUTIONS

Chapter 2, (3.7)

3. Write R^m as a disjoint union of semialgebraic sets $A_0, A_1, \dots, A_k$ with $A_1, \dots, A_k$ connected such that if $x \in A_0$, then $Q(x, T) = 0$, while if $1 \leq i \leq k$ and $x \in A_i$, then the real roots of $Q(x, T)$ are given by continuous semialgebraic functions $\zeta_{i1}(x) < \dots < \zeta_{ie(i)}(x)$ of x . Note that then g must coincide on each A_i ($i > 0$) with one of the functions ζ_{ij} . Next use the fact that each point in A_0 is in the closure of $A_1 \cup \dots \cup A_k$.

Chapter 3, (2.19)

3. Find a partition of C into cells $C_1, \dots, C_k$ and for each $i = 1, \dots, k$ find continuous definable functions $\gamma_{i0}, \dots, \gamma_{ir(i)}: C_i \rightarrow R$ with

$$\alpha|C_i = \gamma_{i0} < \gamma_{i1} < \dots < \gamma_{ir(i)} = \beta|C_i,$$

such that for each cell $\Gamma(\gamma_{ij})$ ($1 \leq j < r(i)$) and each cell $(\gamma_{ij}, \gamma_{ij+1})$ ($0 \leq j < r(i)$):

- (i) the restriction of f to $\Gamma(\gamma_{ij})$ is continuous,
- (ii) the restriction of f to $(\gamma_{ij}, \gamma_{ij+1})$ is continuous, and either strictly increasing in the $(m+1)^{\text{th}}$ variable, or independent of the $(m+1)^{\text{th}}$ variable, or strictly decreasing in the $(m+1)^{\text{th}}$ variable.

Then one shows along the lines of the proof of lemma (2.16) of Chapter 3 that each restriction $f|(\gamma_{i0}, \gamma_{ir(i)})$ is continuous.

Chapter 3, (3.8)

Given an L -formula $\phi(x_1, \dots, x_m, y)$, there are natural numbers M and N such that for all $r \in R^m$ the set $\phi(r, R) \subseteq R$ is a union of at most M intervals and at most N points. This property is inherited by R' .

Chapter 4, (1.17)

3. Use the canonical homeomorphism of A with an open cell to show that a nonempty definable open subset of A has dimension d .

4. Partition A into finitely many cells, and note that $\{a \in \text{cl}(A) : \dim_a(A) < d\}$ is contained in the union of the closures of the cells of dimension $< d$ in this partition.

Chapter 6, (1.15)

2 and 3. With $T \subseteq X$ as in exercise 1, let $f: X \rightarrow T$ be the map assigning to each x its representative in T . Now apply (1.6)(ii) and (2.11) from Chapter 4.

4. If for a certain $\epsilon > 0$ there is no such δ , find definable continuous maps $\gamma_1, \gamma_2: (0, a] \rightarrow X$ for some $a > 0$ such that $|f(\gamma_1(t)) - f(\gamma_2(t))| \geq \epsilon$ for all $t \in (0, a]$, and $\lim_{t \rightarrow 0} |\gamma_1(t) - \gamma_2(t)| = 0$.

5. Consider points in X where the function $x \mapsto |f(x) - x|: X \rightarrow R$ takes its minimum value.

Chapter 6, (4.8)

2. Suppose f is definably proper with respect to $(R, <, \mathcal{S})$, and let α be a curve in X definable in $(R, <, \mathcal{S}')$, such that $f(\alpha) \rightarrow y \in Y$. Since Y is locally closed, y has a neighborhood N in Y that is definable in $(R, <, \mathcal{S})$ and closed and bounded in R^n . To show α is completable in X we may as well assume that $f(\alpha)$ lies entirely in N . Then α lies in $f^{-1}(N)$ and $f^{-1}(N)$ is closed and bounded in R^m , so α is completable in $f^{-1}(N)$, and hence in X .

3. Suppose f is definably proper. Let $S \subseteq X$ be closed in X . To show $f(S)$ is closed in Y , write $S = \bigcap_i S_i$ where $(S_i)_{i \in I}$ is a family of definable closed subsets of X such that for any two indices i_1 and i_2 there is an index i with $S_i \subseteq S_{i_1} \cap S_{i_2}$. It suffices to show that $f(S) = \bigcap_i f(S_i)$. Let $q \in \bigcap_i f(S_i)$, so that $f^{-1}(q) \cap S_i$ is a closed nonempty subset of $f^{-1}(q)$ for all i . Since $f^{-1}(q)$ is compact, it follows that the sets $f^{-1}(q) \cap S_i$ have a common point p , so $p \in S$ and $f(p) = q$.

Chapter 7, (4.3)

1. Let $\phi: R^{m+1} \rightarrow \{x \in R^{m+1} : \|x\| < 1\}$ be the definable homeomorphism given by $\phi(x) = x/\sqrt{1 + \|x\|^2}$. Use that $u \in S^m$ is an asymptotic direction for A if and only if $u \in \text{cl}(\phi(A))$. Then apply theorem (1.8) from Chapter 4 to $\phi(A)$.

3. Use the fact that a basic semilinear set in R^m is an open subset of an affine subspace of R^m , and that R^m cannot be covered by finitely many linear subspaces of dimension $< m$.

Chapter 8, (2.14)

2. (Bounded semilinear sets are polyhedrons.) Along the same lines as the proof of the triangulation theorem. Assume the desired result holds for a certain value of m , and let $S, S_1, \dots, S_k$ be bounded semilinear sets in R^m , $S_i \subseteq S$. Replacing S by its closure (and adding the old S to the list of distinguished subsets) we may as well assume that S is closed. Let $T := \text{bd}(S) \cup \text{bd}(S_1) \cup \dots \cup \text{bd}(S_k)$, so T is closed and bounded of dimension $< m + 1$. After applying a linear automorphism of R^{m+1} we may assume by exercise 3 of Chapter 7, (4.3) that e_{m+1} is a good direction vector for T .

By the inductive hypothesis, and by Chapter 1, (7.4), there are a closed complex K in R^m and a finite collection F of affine functions on R^m , such that $|K| = \pi(T)$, the restrictions to each $\sigma \in K$ of the functions in F can be arranged in increasing order as $f_{\sigma,1} < \dots < f_{\sigma,j(\sigma)}$, and $T = \bigcup \{ \Gamma(f_{\sigma,j}) : \sigma \in K, 1 \leq j \leq j(\sigma) \}$. We also take care that K is compatible with all sets $\pi(S \cap \Gamma(f))$ and $\pi(S_i \cap \Gamma(f))$ ($f \in F, 1 \leq i \leq k$). We now indicate how to use the arguments in the proof of lemma (2.8) to get a complex L in R^{m+1} such that $K = \{ \pi(\sigma) : \sigma \in L \}$, $|L| = S$ and each S_i is a union of simplexes of L .

Indeed, let $A \subseteq R^m$ be any closed and bounded semilinear set, F any finite nonempty collection of affine functions on R^m , and K any closed complex in R^m with $A = |K|$, such that for each $\sigma \in K$ the restrictions of the functions in F to σ can be arranged in increasing order as $f_{\sigma,1} < \dots < f_{\sigma,j(\sigma)}$. (It may happen that different functions in F have the same restriction to σ .)

Given two successive functions $f = f_{\sigma,i}$ and $g = f_{\sigma,i+1}$ we note that for at least one vertex p of σ we must have $\text{cl}(f)(p) < \text{cl}(g)(p)$, where $\text{cl}(f), \text{cl}(g)$ denote the continuous extensions of f and g to $\text{cl}(\sigma)$. Fixing a linear order on $\text{Vert}(K)$ we obtain in this way for each successive pair f, g as above a closed complex $L(f, g)$ in R^{m+1} such that $|L(f, g)|$ is the convex hull of

$$\{ \text{cl}(f)(p) : p \text{ a vertex of } \sigma \} \cup \{ \text{cl}(g)(p) : p \text{ a vertex of } \sigma \},$$

which equals $[\text{cl}(f), \text{cl}(g)]$, and also for each $f = f_{\sigma,i}$ a closed complex $L(f)$ in R^{m+1} with $|L(f)| = \Gamma(\text{cl}(f))$.

All this is just as in the proof of lemma (2.8). Finally, the union of the $L(f, g)$'s and $L(f)$'s is a closed complex in R^{m+1} , and this complex is compatible with the sets in $K^F := \text{set of the } (f, g)\text{'s and } \Gamma(f)\text{'s}$.

3. Let $\mathcal{R}$ be an L -structure, where L extends the language of ordered rings. We may as well assume that S is defined by an L -formula $\phi(x, y)$, $x = (x_1, \dots, x_m)$, $y = (y_1, \dots, y_n)$, since the constants from R that may be involved in an $L(R)$ -formula defining S can be replaced by extra parametric variables (increasing m).

Let $\mathcal{R}'$ be any L -structure elementarily equivalent to $\mathcal{R}$. Then $\mathcal{R}'$ is also o-minimal, by Chapter 3, (3.8), and hence satisfies triangulation. Therefore, given any $a \in R'^m$ there is a definable homeomorphism h_a from $\phi(a, R'^n)$ onto a union of faces of the simplex $(e_1, \dots, e_N)$ in R'^N , for some $N = N(a) \in \mathbb{N}$. Because $\text{Th}(\mathcal{R})$ has definable Skolem functions, we may further assume the graph of h_a is defined by a formula $\psi(a, y, z)$ where $\psi(x, y, z)$ is an L -formula depending on a and with $z = (z_1, \dots, z_N)$. Since $\mathcal{R}' \models \text{Th}(\mathcal{R})$ was arbitrary it follows by model-theoretic compactness that only finitely many different such formulas ψ and numbers $N(a) \in \mathbb{N}$ are needed when $\mathcal{R}'$ varies over all models of $\text{Th}(\mathcal{R})$ and a over R'^m . We can easily construct from these finitely many ψ 's a single definable map f as required, with N the maximum of the finitely many numbers $N(a)$.

